Comparing and combining molecular activities inferred from different omics modalities

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EBI Multi-omics
2-6 March 2026

HEIDELBERG UNIVERSITY HOSPITAL
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The study:
TGF-beta perturbation on a kidney mesenchymal cell line

Article

SOURCE DATA · TRANSPARENT PROCESS · OPEN ACCESS · CHECK FOR UPDATES

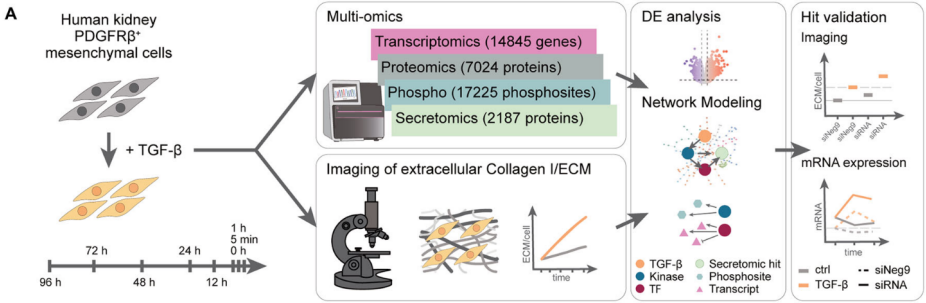
molecular systems biology

Dynamic multi-omics and mechanistic modeling approach uncovers novel mechanisms of kidney fibrosis progression

Nadine Tuechler[1,2,3,4,9], Mira Lea Burtscher[1,3,4,9], Martin Garrido-Rodriguez[1,3,5], Muzamil Majid Khan[1,6], Dénes Türei[3], Christian Tischer[1], Sarah Kaspar[1], Jennifer Jasmin Schwarz[1], Frank Stein[1], Mandy Rettel[1], Rafael Kramann[7,8], Mikhail M Savitski[1,2], Julio Saez-Rodriguez[2,3,5]✉ & Rainer Pepperkok[1,2]✉

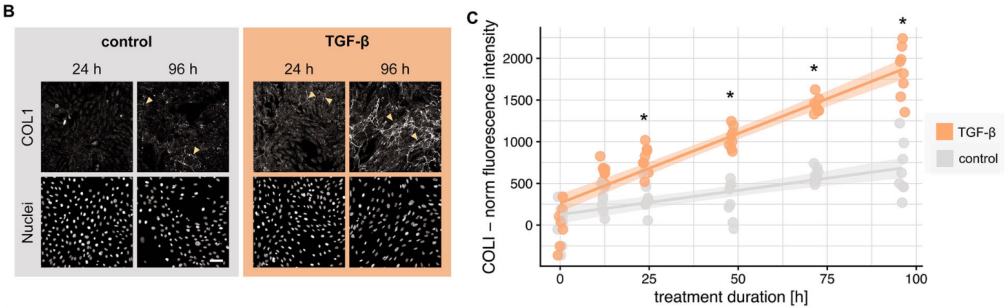
- Tuchler, Burtscher et al., Mol Sys Biol 21(8):1030-1065 (2025).
- Human kidney PDGFRβ+ mesenchymal cell line
- TGF-β stimulation: fibrosis model
- Collagen production: ECM deposition proxy

Overview of the study

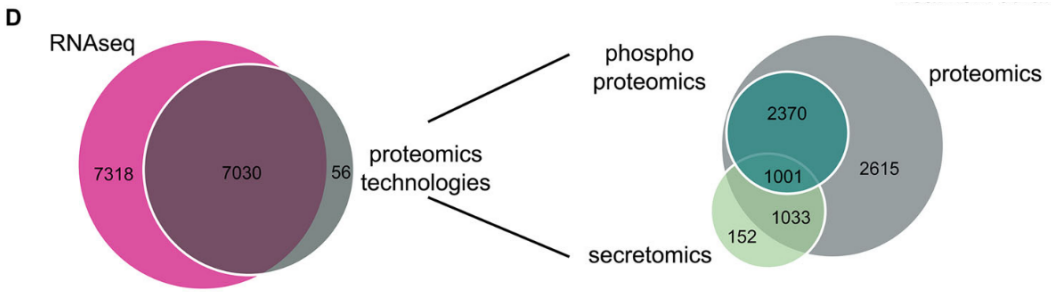


- Longitudinal setup: 7 time points
- Transcriptomics, proteomics, phosphoproteomics, secretomics
- Imaging of collagen
- DE analysis
- Mechanistic modeling

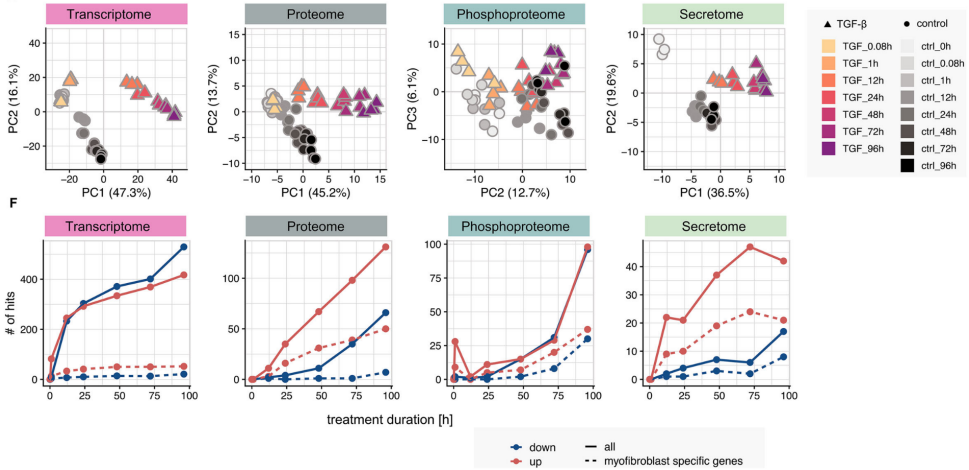
Collagen production upon TGF-beta stimulation



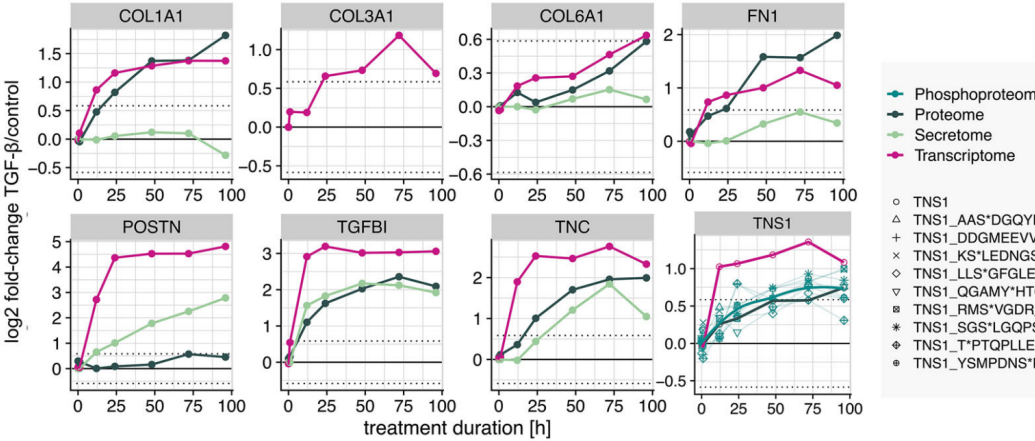
Transcripts and proteins measured



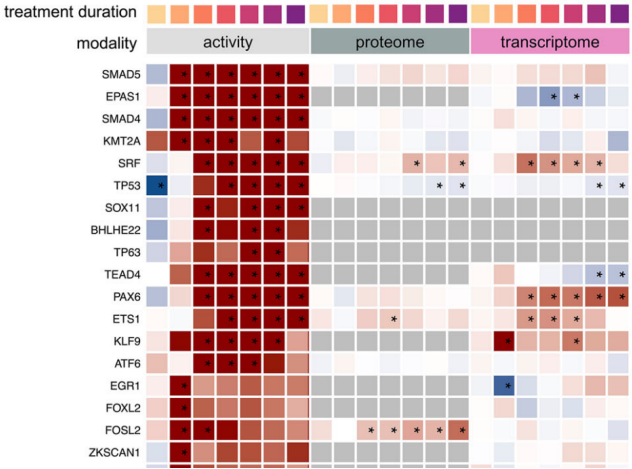
PCA of samples and number of DE transcripts or genes by omics modality



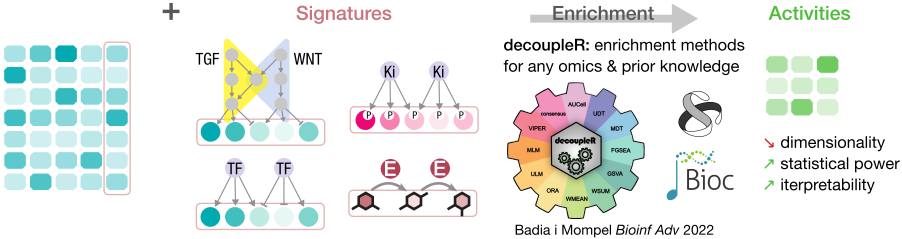
Different time dynamics by omics modality



Diff. exp vs. activities



Activity inference using prior knowledge of omics signatures



Badia i Mompel *Bioinf Adv* 2022

Network modeling by causal reasoning

Activities

Pathway, TF, kinase, metabolic enzyme...

+

Network knowledge

- Causal, direct interactions
- 160k PPI

Turei *Nature Methods* 2016
Turei *Mol Sys Bio* 2021

Optimization

Causal reasoning: CORNETO, CARNIVAL
Boolean modeling: CellNOpt, MaBoSS

Rodriguez-Mier *bioRxiv* 2024
Liu *NPJ Syst Biol Appl* 2019

Context specific networks

- Condition specific models
- Mechanistic insights
- Testable hypotheses

Software tools and environment

- Python with `uv` environment
- OmniPath database access: `omnipath` package
- Activity inference: `decoupler` package
- Causal reasoning: `corneto` package
- Data handling, visualization: `pandas`, `seaborn`

Available datasets

- Differential all omics (transcriptomics, phosphoproteomics, secretomics, proteomics; 7 time points)
- Activities: transcription factor & kinase
- Imaging data: collagen I deposition

Workflow overview - part 1

1. TF activities from transcriptomics
   - Univariate linear model (ulm) by `decoupler`
   - Prior-knowledge: TF regulons from CollecTRI
2. Data and PKN preparation
   - Select the times and modalities
   - Filter to significant changes
   - Causal network from database (OmniPath)
   - Filter the network to expressed genes
   - Prune network to reachable nodes
   - Prune data to network
3. Visualize activities and secretome

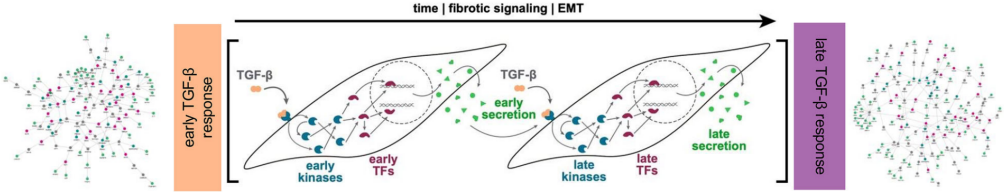
Workflow overview - part 2

1. Infer early network
   - `carnival` causal reasoning implementation by `corneto`
   - Prior-knowledge: human signaling network from OmniPath
   - Perturbations (inputs): early activities + TGF-beta
   - Measurements (outputs): early secretome
2. Infer late network
   - Perturbations (inputs): early secretome + TGF-beta
   - Measurements (outputs): late activities
3. Visualize and compare the context specific networks with activities color coded

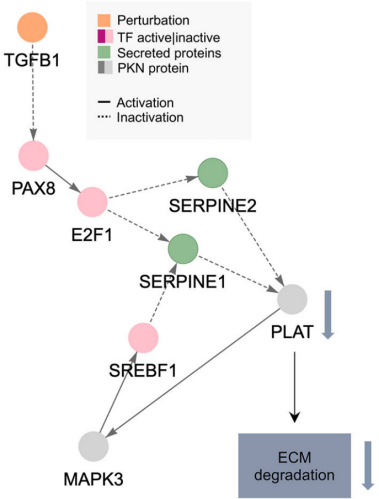
Longitudinal multi-omics

TGFβ $\rightarrow$ early signaling $\rightarrow$ early secretome $\rightarrow$ late signaling $\rightarrow$ late secretome

Each arrow is a network inference



Longitudinal multi-omics



Project materials

https://github.com/saezlab/corneto-ebi-multiomics